Task 1: Linear Regression House Price Predictor

Objective

This report summarizes a complete machine learning workflow using the California Housing dataset. The goal is to predict median district house value from demographic, geographic, and household features using a Linear Regression model.

Dataset Snapshot

The dataset contains 20,640 rows, 8 input features, and one target column. The target is median house value, stored in units of 100,000 USD. The scikit-learn dataset has no missing values.

Modeling Approach

The data was split into 80 percent training and 20 percent testing sets with a fixed random seed for reproducibility. The pipeline standardized features with StandardScaler, then trained a LinearRegression model.

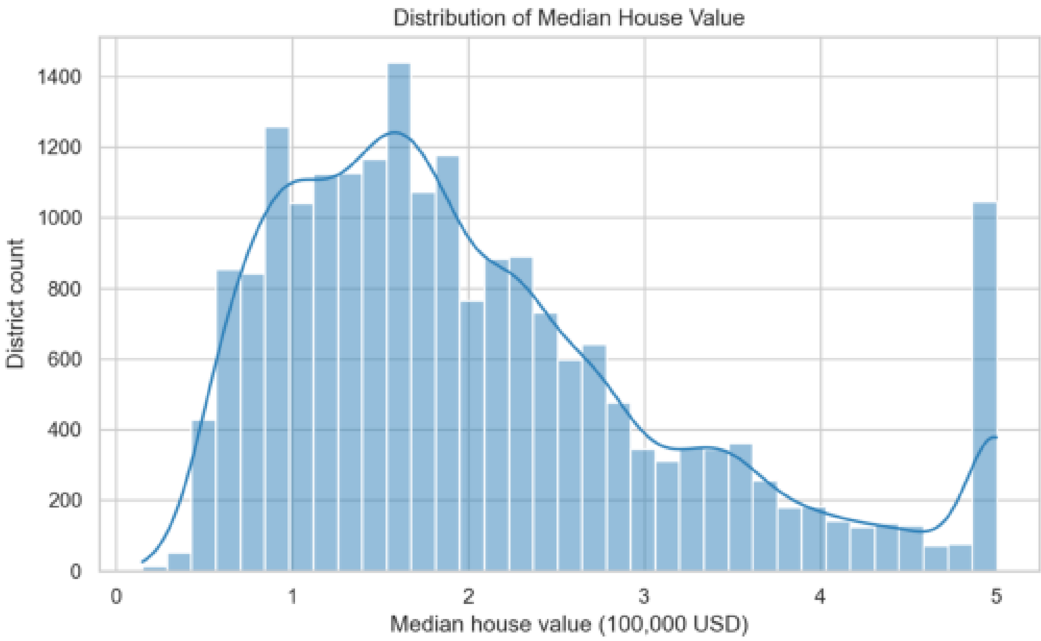
Test Set Metrics

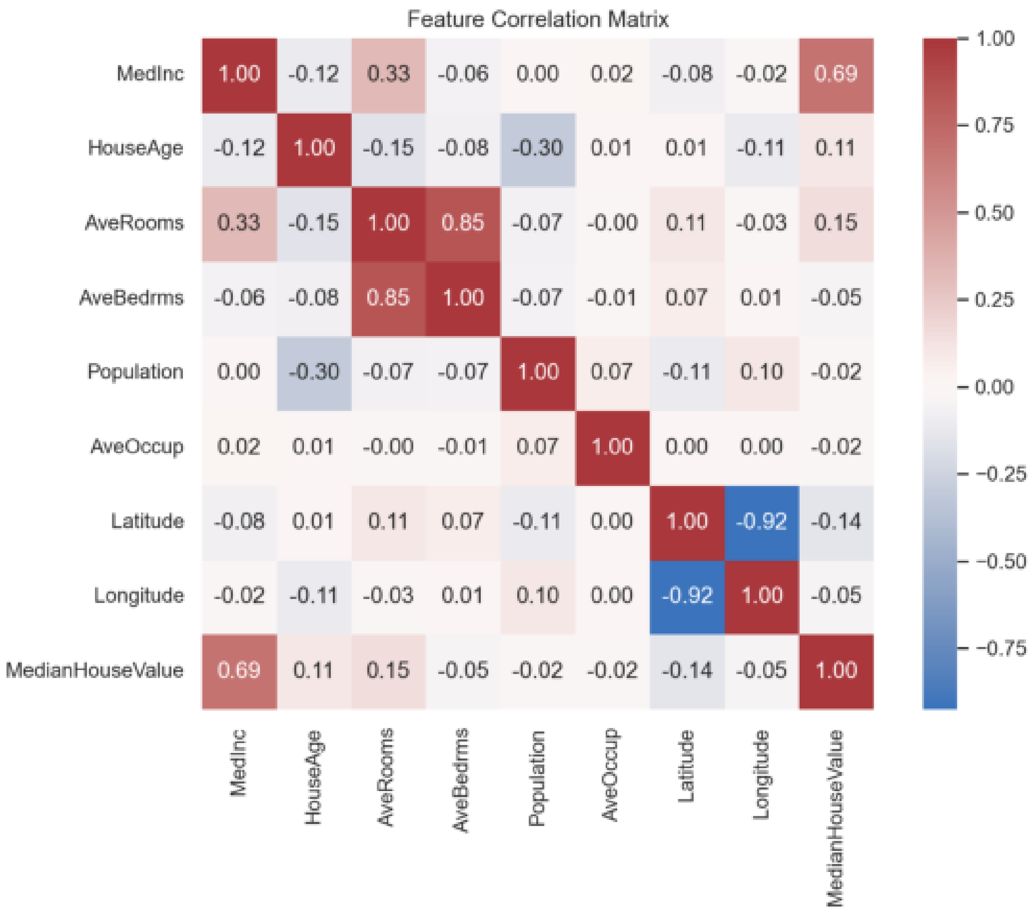
Metric	Value	Interpretation
MAE	0.533	Average absolute error in 100,000 USD units
RMSE	0.746	Penalizes larger prediction errors
R2	0.576	Share of test-set variance explained

The model achieved an R2 of 0.576. This is a useful baseline, but the remaining error indicates that linear relationships alone do not capture all housing price patterns.

Exploratory Data Analysis

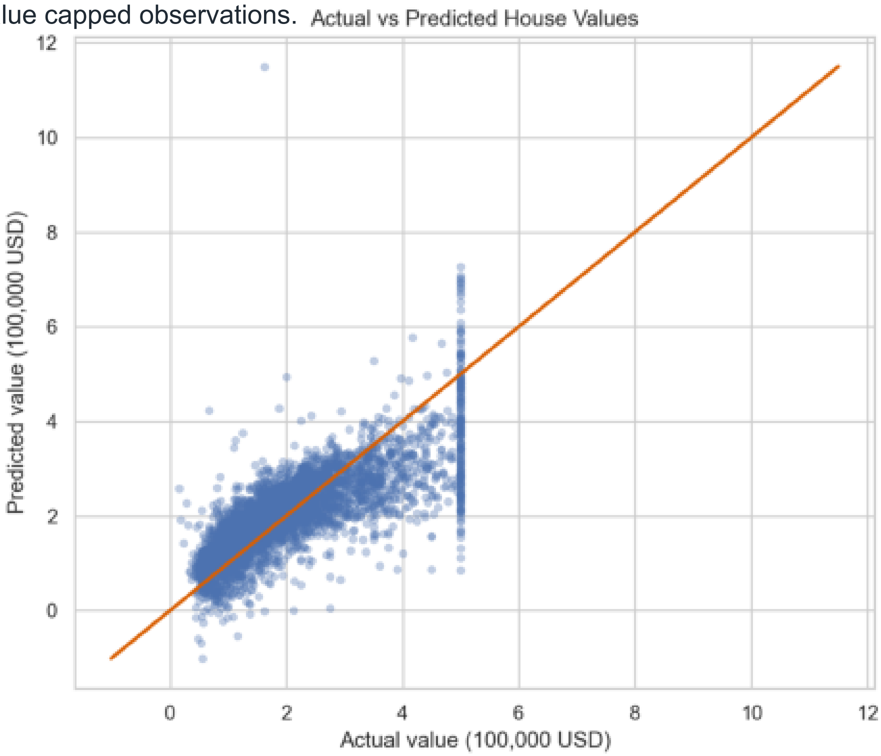
Median income is the strongest positive signal for house value. Location features also matter, and several predictors show correlation, which is expected for geographic housing data.

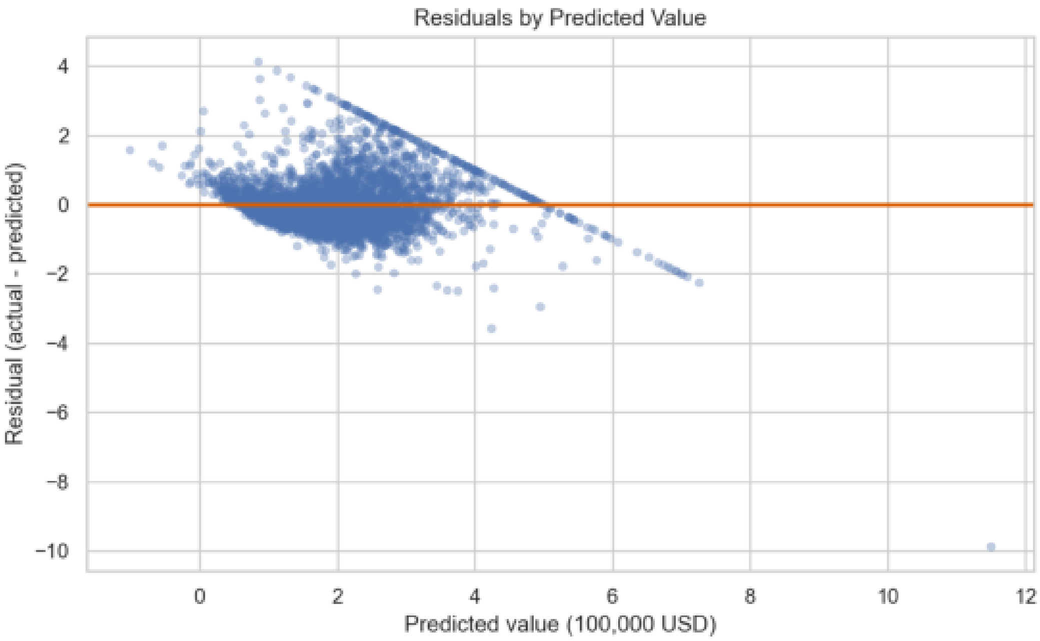




Model Diagnostics

Predicted values generally follow actual values, but the diagnostic plots show spread around the ideal line. This suggests room for non-linear models, feature engineering, and closer handling of high-value capped observations.

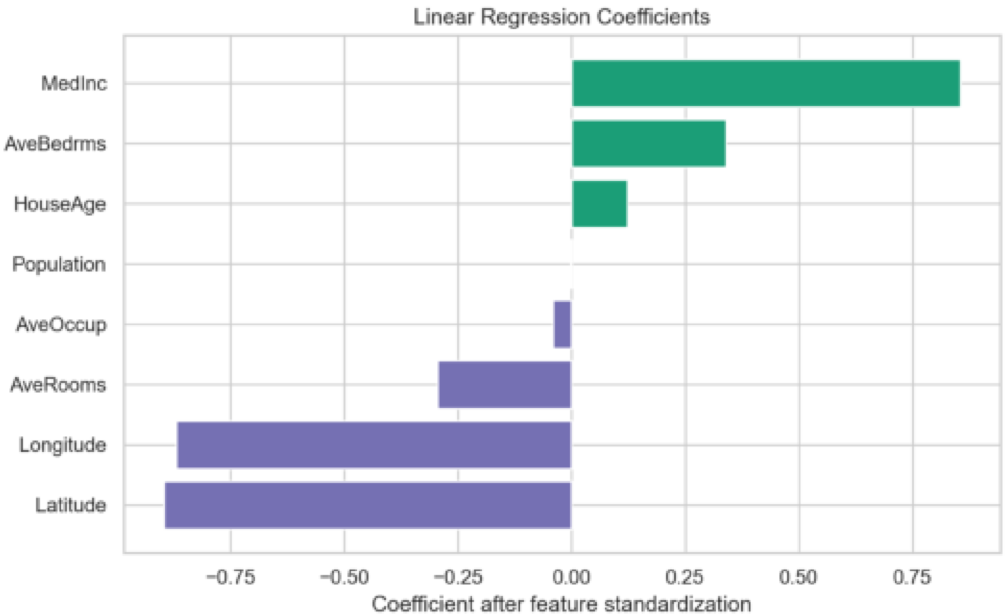




Feature Influence and Improvement Ideas

The five largest standardized coefficients by absolute value are shown below. Coefficients are directional signals for this linear model, not proof of causation.

Feature	Standardized Coefficient
Latitude	-0.897
Longitude	-0.870
MedInc	0.854
AveBedrms	0.339
AveRooms	-0.294



Improvement Ideas

- Test tree-based models such as Random Forest or Gradient Boosting.
- Add engineered features such as rooms per bedroom or location clusters.
- Use cross-validation to estimate performance stability.
- Investigate capped target values and outliers before production use.
- Compare against a simple baseline mean predictor.